

Preliminary Cycle Specification

Night Ravens

Executive Summary

Using previously calculated flight performance and mission parameters, on-design cycle design was conducted with cruise as the design point. The resulting cycle produces 6.10 kN or 1371.35 lb of thrust, approximately 11 lb more than required. The TSFC lower compared to initial estimates at 21.46 g/(kN*s) or 0.7577 lb/(lb*hr), with the initial estimate being 0.867 lb/(lb*hr). The cycle meets the RFP requirement for bleeds totaling 21%. Key station data is provided following SAE ARP 755D numbering. Sensitivity studies were conducted to determine the effects of design choices on specific thrust and specific fuel consumption (SFC).

Attribution & Declaration

The undersigned declare:

- *That all work is presented herein is their own unless otherwise stated.*
- *To have abided by the university honor code and academic integrity policy.*
- *To have read and upheld the NSPE Code of Ethics for Engineers in the production of this work.*
- *That they have reviewed the entirety of the contents presented herein and take full responsibility for all results and data presented.*
- *That all team members agree with the areas of responsibility assigned below.*

Engineer	Area of Responsibility	Signature
John Clardy	GasTurb Manager, Mission Analysis Excel Sheet, Report Writing & Review	<i>John Clardy</i>
Lauren Martins	Preliminary Cycle Analysis Excel Sheet, External Research, Report Writing & Review	<i>Lauren Martins</i>

Introduction

The RFP requests a subsonic jet trainer for both the civilian and defense industries, with civil certification under the Federal Aviation Administration (FAA) and the European Union Aviation Safety Agency (EASA). It is requested that the aircraft be competitive against modern jet trainers. Since this report focuses on cycle analysis, the engines from the AIDC T-5 Brave Eagle – F124 Turbofan Engine – and the Aero L-39 Skyfox – Williams FJ44 – were used as close references.

Table 1: RFP Requirements

Take-Off Runway Length ≤ 2500 ft	Density Altitude 3000 ft	Take-Off Configuration, $C_{L, \max}$ 1.6	Landing Configuration, $C_{L, \max}$ 2.3	Crew Weight 200 kg	Payload Weight 300 kg
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Cruising Time 90 mins	Fuel Reserve Time 45 mins	Service Ceiling ≥ 42000 ft	Cruising Time 90 mins	Front-Face Diameter 25 in	Engine Length 3.4 m
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Core Flow Bleed (Customer) 1%	Core Flow Bleed (Turbine Sealing/Packing) 5%	Core Flow Bleed (Turbine Cooling) 15%	Shaft Power (all Conditions) 100 kW	Turbine Entry Temperature (TET) < 1850 K
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Maneuver 3x360° 3g turns	Mach 0.5	Altitude ≥ 10000 ft
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This report focuses on the content in the third row of Table 1 (along with the engine diameter and length from the second row), which will be used to conduct a preliminary cycle analysis.

Requirements Specification

Based on the preliminary analysis done on the engine and the RFP, the following requirements were found:

Table 2: Engine operation requirements at cruise

Parameters	Values
Maximum Engine Front Face Diameter	25 in
Maximum Engine Length	3.5 m
Required Installed Thrust	1360.47 lbf
Required Specific Thrust	64.1 lbf
Required TSFC	0.867

Methodology

Initially, the cycle analysis was modeled in an Excel Spreadsheet, however minor optimizations – which were conducted entirely in GasTurb – were done prior to writing this report. From this point onwards, GasTurb became the main program used to model the cycle discussed in this report, being deemed amongst group members to be a more accurate modeling of an engine cycle than Excel.

An “on-design” cycle analysis was conducted, with cruise being the design point chosen where the cycle would be analyzed. This point was chosen to optimize fuel consumption (TSFC) [4]. The gas model chosen for this analysis assumed a thermally perfect gas: an ideal gas with properties that vary with temperature. Under this assumption, neither gamma nor heat capacity were considered constant. Component efficiencies and engine Mach values were assumed for simplification using typical values.

Installation effects were also considered for this cycle analysis. The installation effect factor for this engine is 1.05; this value represents the loss that comes from mounting the engine on an aircraft, as it will always produce less thrust when mounted compared to unmounted testing. The value used for the actual installed thrust required from the engine at cruise altitude factors in this installation effect factor, already accounting for the 5% thrust loss that will come from mounting the engine on the jet trainer.

Cycle Design Point

Cruise was chosen as the cycle design point to design a cycle with good TSFC at cruise. This will make a strong foundation for off design for designing a cycle with low TSFC while meeting all the requirements. While this cycle is optimal for cruise performance, further analysis in off design will determine the cycle's viability for other mission design points. The cycle was designed for a two-spool bypass turbofan engine with a mixer and no afterburner. The preliminary cycle was optimized in GasTurb and then sensitivity studies were conducted on the cycle.

Table 3: Sensitivity Study

Study	Base	Delta	SFC (%)	Sp. Thrust (%)
Burner Exit Temperature (K)	1606.68	10	0.43	0.81
Inner Fan Pressure Ratio	2.8	0.1	-0.45	-0.25
Outer Fan Pressure Ratio	2.8	0.1	-0.42	0.42
Design Bypass Ratio	1.42	0.2	-2.67	-5.1
Rel. Overboard Bleed W_Bld/W25	0.01	0.01	0.53	-1.79
HPT NGV 1 Cooling Air / W25	0.05	0.01	-0.42	-0.85
HPT Rotor 1 Cooling Air / W25	0.05	0.01	-0.27	-0.99
LPT Rotor 1 Cooling Air / W25	0.05	0.01	-0.28	-0.99
Rel. HP Leakage to LPT exit	0.05	0.01	-0.16	-1.11
Polytr. Inner LPC Efficiency	0.92	-0.01	0.06	-0.28
Polytr. Outer LPC Efficiency	0.92	-0.01	0.1	-0.1
Polytr. HPC Efficiency	0.88	-0.01	0.15	-0.72
Polytr. HPT Efficiency	0.9	-0.01	0.26	-0.25
Polytr. LPT Efficiency	0.91	-0.01	0.17	-0.17
Polytr. Inner LPC Efficiency	0.92	0.01	-0.05	0.27
Polytr. Outer LPC Efficiency	0.92	0.01	-0.1	0.1
Polytr. HPC Efficiency	0.88	0.01	-0.13	0.68
Polytr. HPT Efficiency	0.9	0.01	-0.24	0.24
Polytr. LPT Efficiency	0.91	0.01	-0.16	0.16
Burner Design Efficiency	0.99	-0.01	0.95	0.06
HP Compressor Pressure Ratio	10.51	0.1	-0.12	-0.08

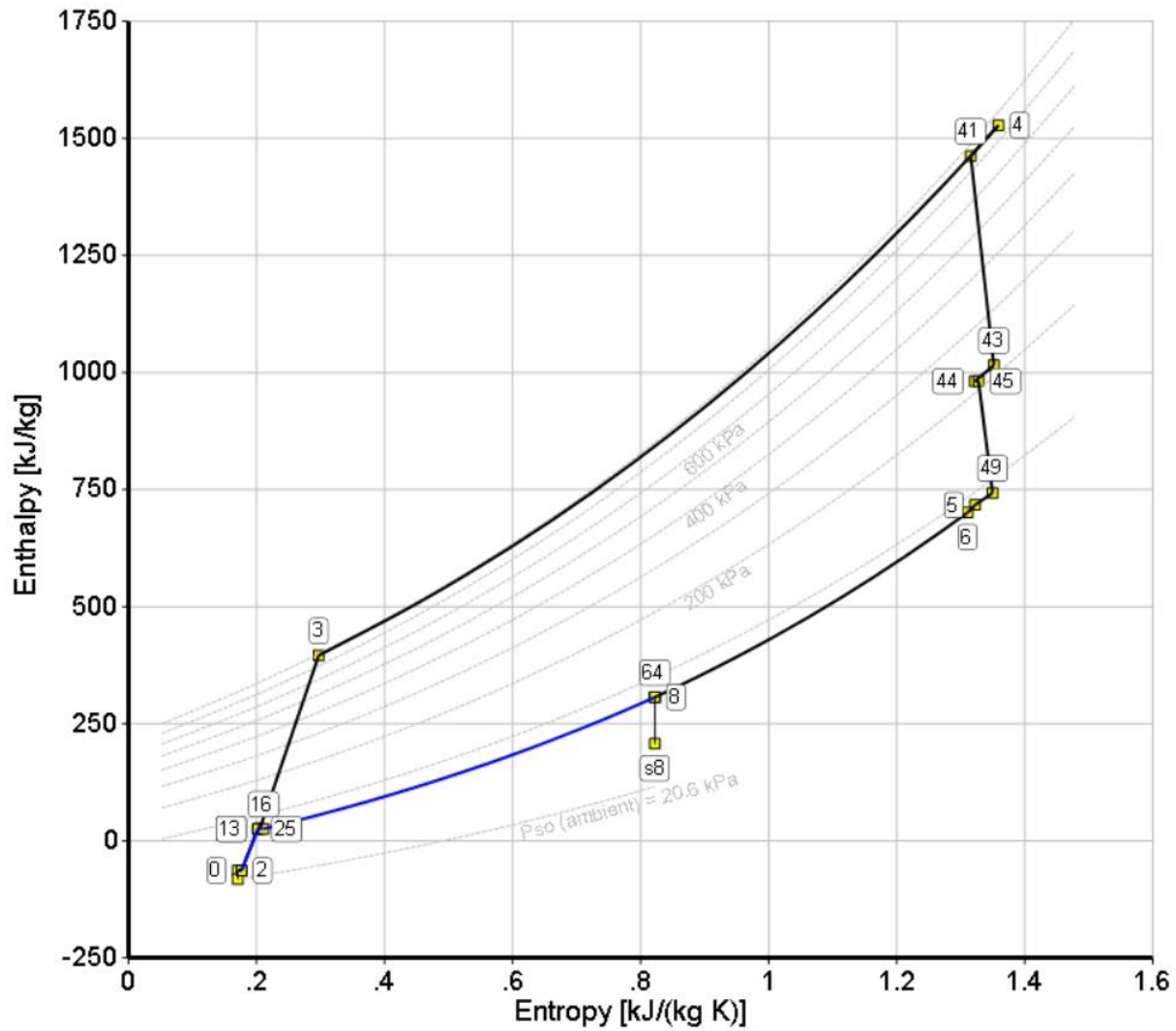


Figure 1: Mollier h-s Diagram

Table 4: Final Selected Cycle Parameters

Parameter	Value
OPR	28.84
TET	1606.68 K
FPR	2.8
BPR	1.42

Table 5: Performance Metrics

Parameter	Value	Value
Thrust	6.10 kN	1371.35 lb
Specific Thrust	415.28 m/s	1362.48 ft/s
TSFC	21.46 g/(kN*s)	0.7577 lb/(lb*hr)

Table 6: Efficiency Assumptions

Parameter	Value
LPC (Fan)	0.92 (Polytropic)
HPC	0.88 (Polytropic)
Burner	0.9995 (Isentropic)
HPT	0.90 (Polytropic)
LPT	0.91 (Polytropic)
Mixer	0.5 (Isentropic)
HP/LP Spool Mechanical	0.99

Table 7: Bleed Flow Assumptions

Parameter	Value
Overboard (Customer) Bleed	1%
LPT Cooling	5%
HP Leak to LPT Exit (Packing)	5%
NGV Cooling	5%
HPT Cooling	5%
Total	21%

Table 8: Thermodynamic properties at major component inlet stations

Parameter	2 Fan Inlet	25 HPC Inlet	3 Burner Inlet	4 HPT Inlet	45 LPT Inlet	13 Bypass Inlet	64 Mixer
Total Temperature (K)	234.997	323.569	682.163	1606.68	1174.73	323.569	595.229
Total Enthalpy (J/kg)	-63357	25625.3	396123	1528220	981613	25625.3	306810
Mass Flow (kg/s)	14.689	6.071	5.767	4.927	5.534	8.618	14.759
Corrected Mass Flow (kg/s)	50	8.747	1.148	1.551	5.358	12.294	27.601

Table 9: Stations with static quantity assumptions

Parameter	1 Inlet Face	2 Fan Inlet	21 / 13 Fan Exit / Bypass Inlet	3 HPC Exit	4 Burner Exit	5 LPT Exit	16 Bypass Exit
Mach	0.65	0.5	0.52	0.55	0.4	0.5	0.5
Area (m ²)	0.235	0.277	0.0468 0.0664	0.00602	0.0106	0.0678	0.0704
Velocity (m/s)	191.90	150.00	182.63	277.26	305.08	295.76	129.54
Static Pressure (kPa)	20.65	22.67	62.61	640.39	686.21	76.64	61.56

Table 10: Turbomachinery Performance

Turbomachine	Corrected Mass Flow (kg/s)	PR / ER	Efficiency (Polytropic)	Power Production (kW)
LPC (Fan Root)	8.660	2.800	0.92	-540.192
LPC (Fan Tip)	12.294	2.800	0.92	-766.863
HPC	1.148	10.514	0.88	-2136.749
HPT	5.358	3.525	0.90	2693.045
LPT	11.939	2.343	0.91	1462.730

References

- [1] “Aero L-39 Skyfox,” Wikipedia, Wikimedia Foundation, 17 Jan. 2026, en.wikipedia.org/wiki/Aero_L-39_Skyfox.
- [2] “AIDC T-5 Brave Eagle,” Wikipedia, Wikimedia Foundation, 26 Jan. 2026, en.wikipedia.org/wiki/AIDC_T-5_Brave_Eagle.
- [3] “F124 Turbofan Engine,” Honeywell Aerospace Technologies, aerospace.honeywell.com/us/en/products-and-services/products/power-and-propulsion/engines/turbofan-engines/f124-turbofan-engine.
- [4] Ferrer-Vidal, L., “On-design” Cycle Analysis of a Turbofan Engine,” AE435 Course Material.
- [5] “Williams FJ44,” Wikipedia, Wikimedia Foundation, 3 Dec. 2025 en.wikipedia.org/wiki/Williams_FJ44
- [6] SAE International, “Aircraft Propulsion System Performance Station Designation and Nomenclature,” SAE Aerospace Recommended Practice ARP755D, SAE International, Warrendale, PA, Feb. 2014. <https://doi.org/10.4271/ARP755D>

Appendices

Appendix A: Preliminary Excel Cycle

Turbofan Cycle Inputs (Mixed exhaust, 2-spool baseline)

Required Inputs (leave blank)		
m0	Freestream Mach (station 0)	0.65
Altitude_m	Geometric altitude [m] for ISA	11582.4
m1	Mach at station 1 (inlet face)	0.65
m2	Mach at station 2 (fan face)	0.5
BPR	Bypass ratio	1.2
FPR	Fan pressure ratio	2.5
2.1	Mach at 2.1 (core fan/LPC TE)	0.52
n_work_split	IPC/HPC work split fraction (0–1)	0.4
m3_1	Mach at 3.1 (HPC discharge diffuser)	0.55
m4	Mach at 4.0 (combustor outlet)	0.4
TET_Tt4_K	Turbine entry total temperature Tt4 [K]	1700
m5	Mach at 5.0 (LPT outlet)	0.5
m15	Mach at 15 (bypass duct exit)	0.5
mdot0_kgps	Total inlet mass flow W0 [kg/s]	50

Assumed / Model Parameters (leave blank)		
Pt2_Pt1	Inlet total pressure recovery Pt2/Pt1	0.98
npLPC	Fan/LPC polytropic efficiency	0.92
npIPC	IPC polytropic efficiency	0.9
w2_6	Swan neck loss coefficient (2.4→2.6 and/or bypass duct)	0.02
Cpr	Pressure recovery factor diffuser (3.0→3.1)	0.6
dPt_Pt_comb	Combustor total pressure loss fraction $\Delta P_t/P_t$	0.02
ncomb	Combustion efficiency	0.99
nmechHP	HP shaft mechanical efficiency	0.99
npHPT	HPT polytropic efficiency	0.9
loss_45_48	Inter-turbine duct PR (4.5→4.8), 1 if none	0.9
Pt6_Pt6id	Mixer total pressure recovery Pt6/Pt6id	0.98
Tref_K	Corrected flow reference Tt [K] (usually 288.15)	288.15
Pref_kPa	Corrected flow reference Pt [kPa] (usually 101.325)	101.325
R_air	Cold section R [J/kg-K]	287.05
hPR	Fuel lower heating value [J/kg]	4315000
pi_HPC	HPC pressure ratio (2.6→3.0)	7
npHPC	HPC polytropic efficiency	0.88

Derived (auto-calculated)		
T0_K	Ambient static temperature [K] (ISA)	216.65
P0_kPa	Ambient static pressure [kPa] (ISA)	20.64582054

Secondary Air System

% core flow

LPT Cooling	0.03	
NGV Cooling	0.05	
HPT Cooling	0.05	
Total SAS [%]		13

	0	1	2	21	13		
	0	1	2	2.1	12.1		
	Ambient conditions	Inlet face	Fan face	Fan root LPC trailing edge	Fan tip trailing edge		
W [kg/s]		14.9883432	14.68857633	6.676625606	8.011950728		
Wcorr [kg/s]		50	50	10.48483212	12.58179855		
Offtake Flow [kg/s]	0	0	0	0	0		
FAR	0	0	0	0	0		
Pt [kPa]	20.64582054	27.43135738	26.88273024	67.20682559	67.20682559		
Tt [K]	216.65	235.0242885	235.0242885	312.6231102	312.6231102		
ht [kJ/kg]	216.8328686	235.243982	235.243982	313.073347	313.073347		
M	0.65	0.65	0.5	0.52	0.52		
V [m/s]	184.2425465	191.8950055	150.0253507	179.5251755	179.5251755		
Ts [K]	199.7110871	216.65	223.7934421	296.5822201	296.5822201		
Ps [kPa]	15.53870374	20.64582054	22.65893084	55.89251361	55.89251361		
rho [kg/m^3]	0.271053525	0.331983045	0.352723591	0.656524561	0.656524561		
cp [J/kg-K]	1001.992163	1002.044054	1002.044054	1004.597891	1004.597891		
gamma	1.401501009	1.40147187	1.40147187	1.400042985	1.400042985		
R [J/kg-K]	287.05	287.05	287.05	287.05	287.05		
power	0			519.6375311	623.5650374		

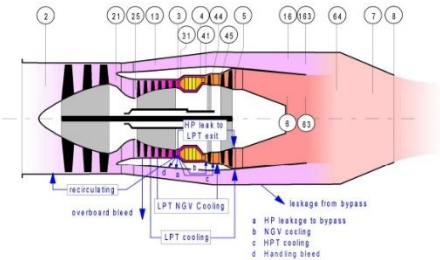
	25		3	4	41		44
	2.6	3	3.1	4	4.05	4.1	4.4
	HPC inlet	HPC outlet	HPC discharge diffuser	Combustor outlet	NGV cool pre-throat	NGV cool post-throat	HPT turbine outlet
W [kg/s]	6.676625606	6.476326838	5.763930886	5.929155627	6.252971969	6.576788	6.576788
Wcorr [kg/s]	10.69880829	2.008796946	1.787829282	3.229691473	3.350533546	3.390767	4.900706
Offtake Flow [kg/s]	0	-0.20029877	0	0	0	0	0
FAR	0	0	0	0.028665288	0.028665288	0.028665	0.028665
Pt [kPa]	65.86268908	461.0388236	461.0388236	451.8180471	451.8180471	451.818	299.9424
Tt [K]	312.6231102	573.9511876	573.9511876	1700	1645.006144	1522.933	1402.009
ht [kJ/kg]	313.073347	579.9066707	579.9066707	1955.605426	1884.363504	1727.358	1573.555
M	0.52	0.55	0.2	0.4	0.4	0.4	0.4
V [m/s]	179.5251755	254.9390821	94.96500055	313.0767901	308.1381588	296.8829	285.9141
Ts [K]	296.5822201	542.8466069	569.6352151	1662.247954	1608.280652	1488.485	1369.149
Ps [kPa]	54.77466334	376.4138503	448.5452064	408.1836132	408.1361656	408.0184	270.6479
rho [kg/m^3]	0.64339407	2.415632791	2.743164418	0.85546577	0.884068915	0.954944	0.688647
cp [J/kg-K]	1004.597891	1044.764698	1044.764698	1298.16908	1292.686901	1279.326	1264.113
gamma	1.400042985	1.378836521	1.378836521	1.283893007	1.285440635	1.289284	1.3
R [J/kg-K]	287.05	287.05	287.05	287.0497162	287.0497162	287.0497	287.0497
power		1781.546202					

	45	5	16	64			8
	4.8	5	15	6	7	8	9
	LPT inlet	LPT outlet	Bypass duct exit	Mixer	Core Nozzle Inlet	Core Nozzle Throat	Core Nozzle Exit
W [kg/s]	6.576788311	6.777087079	8.011950728	14.78903781	14.78903781	14.78904	14.78904
Wcorr [kg/s]	5.445229166	5.551684785	12.83856995	38.64266209	38.64266209	38.64266	38.64266
Offtake Flow [kg/s]	0	0	0	0	0	0	0
FAR	0.028665288	0.028665288	0	0.013135889	0.013135889	0.013136	0.013136
Pt [kPa]	269.9481457	269.9481457	65.86268908	64.5454353	64.5454353	64.54544	64.54544
Tt [K]	1402.008652	1372.491185	312.6231102	798.3084081	798.3084081	798.3084	798.3084
ht [kJ/kg]	1573.554797	1536.300903	313.073347	873.6182434	830.4776036	830.4776	830.4776
M	0.4	0.5	0.5	0.5	0.5	1	1
V [m/s]	285.299458	350.7331903	172.9554469	271.6954076	273.200008	513.3235	513.6866
Ts [K]	1369.813822	1323.678397	297.7347718	765.4026011	763.7705305	678.6543	677.3553
Ps [kPa]	243.6966975	230.273237	55.52324318	54.75609735	54.64652584	34.62181	34.5712
rho [kg/m^3]	0.61977061	0.606043591	0.649662377	0.249221324	0.249254097	0.177723	0.177804
cp [J/kg-K]	1264.112597	1260.056805	1004.597891	1121.66212	1080.527376	1101.095	1090.811
gamma	1.293788375	1.295012975	1.400042985	1.343932011	1.361761824	1.352622	1.357133
R [J/kg-K]	287.0497162	287.0497162	287.05	287.04987	287.04987	287.0499	287.0499
power							

Appendix B: Preliminary GasTurb Cycle

Date: 10Mar26
Time: 21:17 Alt=11562m / M=0.650 ISA

	Units	St 2	St 21	St 25	St 3	St 4	St 44	St 45	St 5	St 6	St 13	St 16	St 64	St 8
Mass Flow	kg/s	14.6889	6.67679	6.67679	6.47648	5.96879	6.63646	6.63646	6.83677	6.83677	8.01214	8.01214	14.8489	14.8489
Total Temperature	K	234.997	313.193	313.193	572.788	1700	1386.5	1386.5	1227.35	1227.35	313.193	313.193	764.567	764.567
Static Temperature	K	223.784	297.169	295.367	568.459	1662.38	1354.43	1354.43	1182.44	1218.69	297.169	298.325	756.489	648.82
Total Pressure	kPa	26.8818	67.2046	66.5326	465.728	451.756	218.339	213.972	128.716	126.142	67.2046	65.1885	75.5778	75.5778
Static Pressure	kPa	22.6592	55.8944	54.1717	453.111	408.122	197.059	193.118	109.666	122.343	55.8944	54.9586	72.5453	40.4064
Velocity	m/s	149.995	179.696	189.488	94.8774	313.063	284.023	284.023	332.885	146.557	179.696	173.12	133.742	503.671
Area	m²	0.277623	0.056705	0.055148	0.024583	0.022292	0.046099	0.04704	0.063565	0.133386	0.068046	0.072113	0.332335	0.135886
Mach Number		0.5	0.52	0.55	0.2	0.4	0.4	0.4	0.5	0.217	0.52	0.5	0.247	1
Density	kg/m³	0.352742	0.655249	0.638928	2.77682	0.855284	0.506863	0.496726	0.323102	0.34973	0.655249	0.641781	0.334081	0.216956
Spec Heat @ T	J/(kg·K)	1003.24	1005.95	1005.95	1045.15	1307.72	1262.92	1262.92	1236.73	1236.73	1005.95	1005.95	1110.65	1110.65
Spec Heat @ Ts	J/(kg·K)	1002.97	1004.73	1004.69	1044.22	1303.75	1258.29	1258.29	1229.05	1235.32	1004.73	1004.76	1108.61	1081.34
Enthalpy @ T	J/kg	43357	15161.3	15161.3	280494	1.66352E6	1.2531E6	1.2531E6	1.05279E6	1.05279E6	15161.3	15161.3	492911	492911
Enthalpy @ Ts	J/kg	-74606.3	-984.11	-2791.61	275994	1.61451E6	1.21276E6	1.21276E6	997388	1.04206E6	-984.11	176.036	483967	366068
Entropy Function @ T		-0.831934	0.17276	0.17276	2.31759	6.93806	5.9958	5.9958	5.45862	5.45862	0.17276	0.17276	3.44163	3.44163
Entropy Function @ Ts		-1.00282	-0.011518	-0.032773	2.25012	6.83648	5.89325	5.89325	5.29844	5.42804	-0.011518	2.0564E-3	3.440068	2.81546
Energy	J/kg	17150.1	90170.8	89545.7	342508	1.43636E6	1.03931E6	1.03805E6	839549	838293	90170.8	88276.5	371958	371958
Gas Constant	J/(kg·K)	287.05	287.05	287.05	287.05	287.045	287.045	287.045	287.045	287.045	287.05	287.05	287.048	287.048
Fuel-Air-Ratio		0	0	0	0	0.033481	0.030011	0.030011	0.029106	0.029106	0	0	0.013194	0.013194
Water-Air-Ratio		0	0	0	0	0	0	0	0	0	0	0	0	0



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GasTurb

Date: 10Mar26
Time: 21:17

mixed Turbofan
Alt=11582m / Mn=0.650 ISA

Invalid Result:

Your input data do not lead to a reasonable cycle. Please check and correct your data!

Station	W kg/s	T K	P kPa	WRstd kg/s
amb		216.65	20.646	
1	14.689	235.00	27.430	
2	14.689	235.00	26.882	50.000
13	8.012	313.19	67.205	12.594
21	6.677	313.19	67.205	10.495
25	6.677	313.19	66.533	10.601
3	6.476	572.79	465.728	1.987
31	5.775	572.79	465.728	
4	5.969	1700.00	451.756	3.252
41	6.303	1647.01	451.756	3.380
43	6.303	1424.82	218.339	
44	6.636	1386.50	218.339	
45	6.636	1386.50	213.972	6.894
49	6.636	1247.67	128.716	
5	6.837	1227.35	128.716	11.107
6	6.837	1227.35	126.142	
16	8.012	313.19	65.188	
64	14.849	764.57	75.578	
8	14.849	764.57	75.578	32.428
Bleed	0.033	572.79	465.727	

Efficiencies:	isent	polytr	RNI	P/P
Outer LPC	0.9000	0.9120	0.337	2.500
Inner LPC	0.9000	0.9120	0.337	2.500
HP Compressor	0.8800	0.9073	0.595	7.000
Burner	0.9995			0.970
HP Turbine	0.9000	0.8924	0.581	2.069
LP Turbine	0.9100	0.9051	0.336	1.662
Mixer	0.5000			

HP Spool mech Eff	1.0000	Nom Spd	22800 rpm
LP Spool mech Eff	1.0000	Nom Spd	14600 rpm

P2/P1= 0.9800 P25/P21=0.9900 P45/P44= 0.9800

hum [%]	war0	FHV	Fuel
0.0	0.00000	43.150	Generic

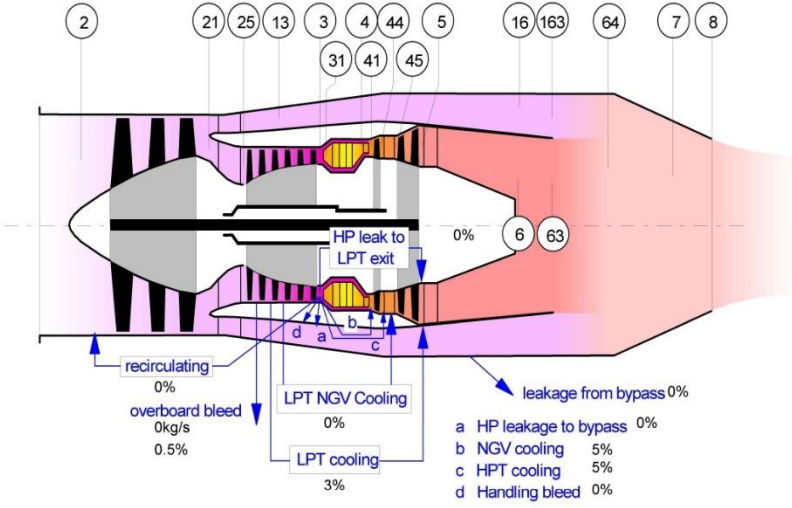
FN = 7.39 kN

TSFC = 26.1747 g/(kN*s)
WF Burner= 0.19337 kg/s
s NOX = 0.1457
BPR = 1.2000
Core Eff = 0.4889
Prop Eff = 0.4365
P3/P2 = 17.325

P16/P6 = 0.51679
A63 = 0.04870 m²
A163 = 0.28352 m²
A64 = 0.33222 m²
XM63 = 1.05004
XM163 = 0.11183
XM64 = 0.24709
P63/P6 = 0.99000
P163/P16 = 0.99000
A8 = 0.14304 m²
CD8 = 0.95000
Ang8 = 25.00 °
P8/Pamb = 3.66062
WLkBy/W25= 0.00000
WCHN/W25 = 0.05000
WCHR/W25 = 0.05000
Loading = 100.00 %
WCLN/W25 = 0.00000
WCLR/W25 = 0.03000
WBHD/W21 = 0.00000
far7 = 0.01319
WBLD/W25 = 0.00500
PWX = 50.0 kW
P16/P13 = 0.9700
P6/P5 = 0.9800

Input Data File:

C:\Users\CLARDYJ\Downloads\prelim.CYM



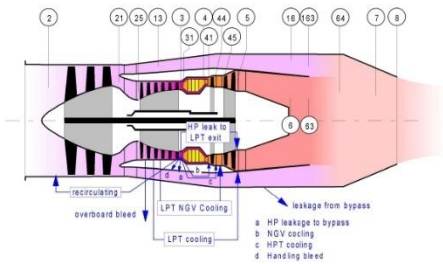
MFanLBC.WMF

GasTurb

Appendix C: Optimized GasTurb Cycle

Date: 08Mar26
Time: 23:26 Alt=11582m / M=0.650 ISA

	Units	St 2	St 21	St 25	St 3	St 4	St 44	St 45	St 5	St 6	St 13	St 16	St 64	St 8
Mass Flow	kg/s	14.6889	6.07078	6.07078	5.76724	4.92684	5.53392	5.53392	5.83746	6.14099	8.61815	8.61815	14.7591	14.7591
Total Temperature	K	234.997	323.569	323.569	682.163	1606.68	1174.73	1174.73	954.849	942.152	323.569	323.57	595.229	595.229
Static Temperature	K	223.784	307.033	305.18	646	1570.31	1146.29	1146.29	917.496	934.974	307.033	308.222	568.573	500.798
Total Pressure	kPa	26.8818	75.2692	74.5165	783.453	759.949	215.599	211.287	90.1844	88.3807	75.2692	73.0111	77.8728	77.8728
Static Pressure	kPa	22.6592	62.6052	60.6757	640.392	686.208	194.375	190.487	76.643	85.6719	62.6052	61.5669	65.8463	41.3719
Velocity	m/s	149.995	182.63	192.589	277.262	305.082	262.733	262.733	295.758	129.541	182.63	175.942	236.756	445.611
Area	m²	0.277623	0.046796	0.045511	6.0231E-3	0.010608	0.036656	0.036383	0.067822	0.148507	0.066431	0.070403	0.154515	0.115085
Mach Number		0.5	0.52	0.55	0.55	0.4	0.4	0.4	0.5	0.217	0.52	0.5	0.5	1
Density	kg/m³	0.352742	0.710342	0.692629	3.45347	1.52237	0.590733	0.578919	0.291016	0.319217	0.710342	0.695753	0.40345	0.287798
Spec Heat @ T	J/(kg·K)	1003.24	1006.85	1006.85	1070.64	1283.69	1218.09	1218.09	1171.94	1167.08	1006.85	1006.85	1062.07	1062.07
Spec Heat @ Ts	J/(kg·K)	1002.97	1005.41	1005.35	1061.99	1279.65	1213	1213	1163.56	1165.48	1005.41	1005.52	1056.02	1040.64
Enthalpy @ T	J/kg	43367	25625.3	25625.3	396123	1.52822E6	981613	981613	717292	701418	25625.3	25625.8	306810	306810
Enthalpy @ Ts	J/kg	-74606.3	8948.45	7080.12	357686	1.48168E6	947099	947099	673556	693027	8948.45	10148	278783	207525
Entropy Function @ T		-0.831934	0.287217	0.287217	2.96073	6.6273	5.23888	5.23888	4.36877	4.30831	0.287217	0.287222	2.48025	2.48025
Entropy Function @ Ts		-1.00282	0.102995	0.081741	2.7591	6.52522	5.13525	5.13525	4.20607	4.27718	0.102995	0.116573	2.3125	1.84778
Exergy	J/kg	17150.1	100565	99939.6	450486	1.35272E6	814102	812845	549688	536314	100565	98670.5	247496	247496
Gas Constant	J/(kg·K)	287.05	287.05	287.05	287.05	287.046	287.046	287.046	287.046	287.047	287.05	287.05	287.049	287.049
Fuel-Air-Ratio		0	0	0	0	0.027298	0.024231	0.024231	0.022942	0.021783	0	0	8.9498E-3	8.9498E-3
Water-Air-Ratio		0	0	0	0	0	0	0	0	0	0	0	0	0



BPALBCWBP

GasTurb

Date: 09Mar26
Time: 14:04

mixed Turbofan
Alt=11582m / Mn=0.650 ISA

Station	W kg/s	T K	P kPa	WRstd kg/s
amb		216.65	20.646	
1	14.689	235.00	27.430	
2	14.689	235.00	26.882	50.000
13	8.618	323.57	75.269	12.294
21	6.071	323.57	75.269	8.660
25	6.071	323.57	74.516	8.747
3	5.767	682.16	783.453	1.148
31	4.796	682.16	783.453	
4	4.927	1606.68	759.949	1.551
41	5.230	1557.99	759.949	1.622
43	5.230	1201.10	215.599	
44	5.534	1174.73	215.599	
45	5.534	1174.73	211.287	5.358
49	5.534	975.81	90.184	
5	5.837	954.85	90.184	11.939
6	6.141	942.15	88.381	
16	8.618	323.57	73.011	
64	14.759	595.23	77.873	
8	14.759	595.23	77.873	27.601
Bleed	0.061	682.16	783.453	

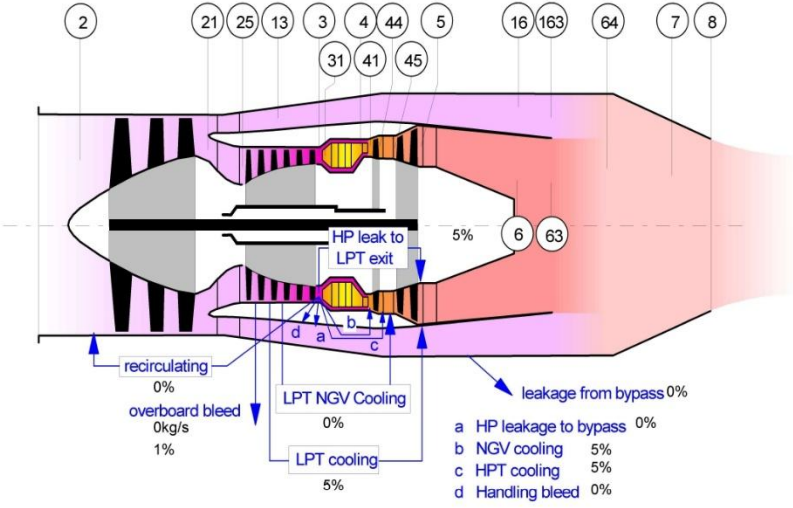
Efficiencies:	isent	polytr	RNI	P/P
Outer LPC	0.9077	0.9200	0.337	2.800
Inner LPC	0.9077	0.9200	0.337	2.800
HP Compressor	0.8375	0.8800	0.641	10.514
Burner	0.9995			0.970
HP Turbine	0.9126	0.9000	1.044	3.525
LP Turbine	0.9181	0.9100	0.402	2.343
Mixer	0.5000			

HP Spool mech Eff	0.9900	Nom Spd	22800 rpm
LP Spool mech Eff	0.9900	Nom Spd	14600 rpm

P2/P1=	0.9800	P25/P21=	0.9900	P45/P44=	0.9800
hum [%]	war0	FHV	Fuel		
0.0	0.00000	43.150	Generic		

FN	=	6.10 kN
TSFC	=	21.4619 g/(kN*s)
WF Burner	=	0.13092 kg/s
s NOX	=	0.3149
BPR	=	1.4196
Core Eff	=	0.4674
Prop Eff	=	0.4820
P3/P2	=	29.144
P16/P6	=	0.82610
A63	=	0.06136 m ²
A163	=	0.09316 m ²
A64	=	0.15451 m ²
XM63	=	0.65762
XM163	=	0.35537
XM64	=	0.50000
P63/P6	=	0.99000
P163/P16	=	0.99000
A8	=	0.12114 m ²
CD8	=	0.95000
Ang8	=	25.00 °
P8/Pamb	=	3.77179
WLkBy/W25	=	0.00000
WCHN/W25	=	0.05000
WCHR/W25	=	0.05000
Loading	=	100.00 %
WCLN/W25	=	0.00000
WCLR/W25	=	0.05000
WBHD/W21	=	0.00000
far7	=	0.00895
WBLD/W25	=	0.01000
PWX	=	110.0 kW
P16/P13	=	0.9700
P6/P5	=	0.9800

Input Data File:
C:\Users\CLARDYJ\Desktop\gasturb\jet_optimized_1.CYM



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GasTurb